Consensus Onset Mechanics, Timestamp Resolution, and Workload Constraints in Sepsis Early-Warning Benchmarks: An Operational Evaluation Audit

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Abstract

Objectives: To evaluate how surveillance coverage, clinical-process anchors, and alert workload constraints govern estimated performance and evaluability in sepsis early warning.

Materials and Methods: Using MIMIC-III as a reproducible benchmark substrate spanning an inter-era EHR vendor transition from CareVue (2001–2008 development) to MetaVision (2008–2012 prospective-replay audit; 8,254 first stays; 97,700 windows; prevalence 0.694%), we evaluated 8 predictive architectures during hours 19–30. We formalized an operational cascade ($S \to V \to O \to D$), quantified order date coarsening, and evaluated alert frontiers under 12-hour alert suppression.

Results: Among 408 operational Sepsis-3 stays ($S = 1$), 257 (63.0%) developed onset after hour 36 ($V = 0$). Among 151 covered stays ($V = 1$), 100 (66.2%) were first-bin assigned ($T_{\text{susp}} - T_{\text{S3}} \in [24\text{h}, 25\text{h})$), structurally guaranteeing pre-suspicion eligibility; 50 of the remaining 51 were also eligible ($O = 1$: 150/151, 99.3%). Conditional on fixed operational Sepsis-3 timestamps, evaluating opportunity against earliest prescription dates under midnight assignment eliminated all horizon-positive observations (0/151 definite; bounds [0.0%, 31.1%] covered, [0.0%, 11.5%] cohort-wide), preventing discrimination estimation. At 10 alert episodes/100 patient-days (407 emitted alerts), BiMamba SSM and LightGBM each detected 7 distinct patients ($D = 1$), yielding episode PPV 1.72% (7/407, NNE 58.1), covered sensitivity 4.64% (7/151), and population yield 1.72% (7/408).

Discussion: Evaluability and lead-time estimates depend fundamentally on protocol coverage and process-anchor definitions.

Conclusion: Sepsis predictive evaluations must report protocol coverage, distinguish structurally induced eligibility from biological lead time, and benchmark under explicit bedside workload constraints.

Keywords: Sepsis Early Warning, Surveillance Truncation, Anchor Selection, Alert Budget Frontier, State-Space Models.

Lay Summary: Machine learning systems designed to provide early warnings for sepsis in intensive care units are frequently evaluated using historical health records that assume alarms could have alerted doctors hours before treatment began. In this study, we audited 8,254 ICU stays and eight AI architectures to examine how real-world clinical timelines affect early warning accuracy. In our implementation, many episodes were assigned an onset roughly 24 hours before the selected suspicion timestamp because of the retrospective labeling rules rather than earlier detection of physiological deterioration. Furthermore, when medication records only record calendar dates rather than minute-by-minute administration, determining whether an alarm arrived before treatment becomes highly uncertain. Finally, when we limited alerts to a prespecified low-volume scenario (10 alert episodes per 100 patient-days), the tested models detected fewer than 2% of incident cases under the restricted monitoring schedule, equivalent to approximately 58 simulated alert episodes per timely detected case. These results highlight the critical need to evaluate healthcare AI under realistic clinical workflows and bedside workload limits.

1 Introduction

1.1 Background and Significance

Sepsis remains a leading cause of critical illness, inpatient mortality, and healthcare expenditure globally [1, 2]. Clinical treatment urgency depends critically on presentation: Surviving Sepsis Campaign guidelines recommend immediate antimicrobial therapy (ideally within 1 hour) for septic shock, and rapid diagnostic assessment for sepsis without shock [3, 4]. To support prompt intervention, informatics research has developed numerous machine learning (ML) early-warning models, frequently reporting AUROC values between 0.80 and 0.88 ahead of consensus onset [5]. However, published ICU prediction studies differ substantially in outcome definitions, temporal alignment, and evaluation designs; reported AUROCs should not be interpreted as directly comparable measures of deployment readiness, even across deep learning architectures evaluated in multi-center international cohorts [6] or broad methodological reviews [7]. Prior MIMIC-III analyses demonstrate that operational onset definitions can shift measured performance more than changing model class [8]. Time-sensitive sepsis benchmarks have also incorporated utility functions that reward and penalize alerts by timing [9], reinforcing the need to report explicit temporal evaluation rules.

Clinical evaluations report heterogeneous performance [10, 11], while clinician-vignette studies show discrepancies between operational onset labels and recommended treatment timing [12]. External validation of the original Epic Sepsis Model demonstrated that hospitalization-level and prediction-level workload estimates are not interchangeable, emphasizing the need to declare the exact evaluation unit [10]. Prospective evaluations of early-warning systems such as TREWS found associations between timely alert confirmation and improved outcomes among patients identified prior to antibiotics [11, 13]; however, observational clinician responses support potential clinical utility without establishing randomized causal efficacy [11]. Furthermore, ICU clinicians already encounter substantial alarm exposure [14]; repeated nonactionable alerts contribute to alert fatigue and lower adoption [15, 16].

Recent evaluations of sepsis early warning before clinical intervention present divergent findings. Kamran et al. [17] reported a severe decline in Epic model AUROC (0.62 to 0.47) when excluding predictions following clinical recognition. Conversely, Dutta et al. [18] demonstrated preserved discrimination under certain pre-clinical action definitions across nine hospitals. Do et al. [19]

further showed that continuous surveillance and alert silencing rules fundamentally alter clinical workload across health systems. Addressing label contamination and circularity in sepsis definitions, Schamoni et al. [20] emphasized that clinical recognition anchors require explicit separation from physiological deterioration. Furthermore, clinical database timestamps carry inherent recording latency, clock drift, and temporal coarsening [21]. This literature indicates that pre-intervention performance does not follow a uniform collapse, but depends on how onset labels, clinical-process anchors, and evaluation units are constructed. Crucially, auditing published benchmarks requires examining the open, schema-documented computational substrate on which consensus phenotyping rules and prediction pipelines are routinely reproduced. We therefore evaluate MIMIC-III not to propose a deployable predictive model, but to isolate how retrospective onset conventions, timestamp resolution, and surveillance eligibility shape apparent early warning before cross-institutional variance is introduced.

These findings motivate the central question of this study: *Why do estimates of pre-clinical action opportunity and alert performance diverge across evaluation designs?* We formalize an operational evaluation audit framework across four distinct cascade stages ($S \rightarrow V \rightarrow O \rightarrow D$):

1. **Incident Target Sepsis (S):** Study-defined operational Sepsis-3 onset ($S = 1$).
2. **Surveillance Protocol Coverage (V):** Scheduled monitoring intervals, showing that 63.0% of episodes develop onset after surveillance ends.
3. **Clinical-Process Anchors (O):** Evaluability relative to clinical suspicion (T_{susp}) vs. medication order proxies ($T_{\text{Rx,start}}$), proving that high pre-suspicion opportunity (150/151, 99.3%) is largely an algebraic artifact of the discrete 24-hour lookback in first-bin assigned cases (100/151, 66.2%).
4. **Emitted-Alert Workload Constraints (D):** Operational alert budgets (5, 10, 20 episodes/100 patient-days) with 12-hour suppression, including a maximum observed population yield of 1.72% at 10 episodes/100 surveillance patient-days.

2 Materials and Methods

2.1 Study Design and Cohort Definition

We structured development and evaluation as an inter-era EHR vendor-transition audit using MIMIC-III (v1.4) [22]: CareVue-era candidate stays (2001–2008; $N = 12,964$ stays; 10,371 train, 2,593 validation; 1,687 incident sepsis) supported upstream development, whereas MetaVision-era stays (2008–2012; $N = 8,254$ analyzed first adult ICU stays; 9,732 candidates; 1,886 sepsis) supported prospective-replay auditing across documentation and timestamping transitions within one institution. Adult first ICU stays with LOS $\geq$ 30h were included. Conditioning on LOS $\geq$ 30h ensures that included patients remain hospitalized through mid-stay surveillance (hours 19–30), but at surveillance entry ($t = 19$h) it retrospectively conditions on 11 hours of future ICU survival. Stays with community-onset sepsis ($T_{S3} \leq 0$h) or developing sepsis during the pre-surveillance baseline ($0 < T_{S3} \leq 19$h, $N = 1,478$) were excluded, yielding the 8,254 test stays (408 incident sepsis, 7,846 controls). Beyond the primary MetaVision evaluation, historical sensitivity analyses evaluated $N = 3,581$ incident sepsis stays partitioned into CareVue-only (1,687), MetaVision-only (1,886), and transitional stays bridging both database systems (8), where $1,687 + 1,886 + 8 = 3,581$ stays form a mutually exclusive, exhaustive partition. Reporting follows TRIPOD+AI and STROBE recommendations [23, 24].

Surveillance was evaluated under a prespecified second-day protocol: predictions were generated at 12 scheduled hourly decision bins $[t-1, t]$ for $t \in \{19, 20, \ldots, 30\}$ hours post-admission (4,070.8 patient-days of surveillance volume). Stays were censored at Sepsis-3 onset ($N = 110$ stays, 623 windows) or EHR telemetry cessation ($N = 180$ control stays, 725 windows). Cohort derivations, exclusions, feature dictionaries, and executable definitions are detailed in the Supplementary Methods.

2.2 Outcome Definition and Operational Clocks

Sepsis was defined according to a study-specific operational adaptation of Sepsis-3 criteria [1, 2]: an acute increase in Sequential Organ Failure Assessment (SOFA) score ≥ 2 points over baseline. In the frozen study implementation, baseline SOFA was defined as the minimum total SOFA observed in the 24-hour interval preceding the selected suspicion anchor ($t \in [T_{\text{susp}} - 24\text{h}, T_{\text{susp}}]$, assumed 0 if unmeasured). Sepsis-3 onset was identified when total SOFA increased by ≥ 2 points above $\text{SOFA}_{\text{base}}$ within the clinical search window $[T_{\text{susp}} - 24\text{h}, T_{\text{susp}} + 12\text{h}]$ relative to suspected infection (defined as blood culture specimen collection paired with systemic antimicrobial prescription orders; detailed in Supplementary Section S1.5). Because $\text{SOFA}_{\text{base}}$ is determined retrospectively across the candidate lookback, for first-bin episodes near $T_{\text{susp}} - 24$h, the minimum can occur after assigned onset if organ dysfunction subsequently resolves; this identifies the first qualifying bin relative to a retrospective search-window minimum rather than an observed increase from a pre-ICU baseline; an alternative antecedent-baseline sensitivity analysis was not performed. In the MIMIC-III database, antimicrobial timing derives from recorded prescription scheduling dates (`prescriptions.startdate`), which represent medication order records rather than electronically verified bedside administration timestamps. While MetaVision records minute-resolved intravenous infusions in `inputevents_mv`, we intentionally audit the date-coarsened `prescriptions` table without constructing an ad-hoc composite anchor: order records reflect standard benchmark practice, preserve compatibility with CareVue cohorts, capture oral and ward orders omitted from continuous infusion tables, and isolate the specific timestamp uncertainty intrinsic to retrospective order records.

To resolve temporal precision and event ambiguity without conflating orders, intended start dates, and physical administrations, we track six operational milestones (with ICU admission as the reference origin $t = 0$, followed by five operational event clocks; Figure 1):

- $T_{\text{admit},i}$: ICU admission timestamp ($t = 0$, reference origin).
- $T_{\text{Rx,start},i}$: Earliest documented systemic antimicrobial order start date across the hospital stay (`prescriptions.startdate`, calendar date coarsened to midnight 00:00:00; captures admission empiric or prophylactic orders).
- $T_{\text{abx,pair},i}$: Systemic antimicrobial order date participating in the qualifying infection-suspicion pair with blood culture ([−24h, +72h] window per Seymour et al. [2]).
- $T_{\text{culture,pair},i}$: Blood culture specimen collection timestamp (`microbiologyevents.charttime`, minute-level precision).
- $T_{\text{susp},i}$: Operational infection-suspicion anchor, defined per Seymour et al. [2] as $\min(T_{\text{abx,pair}}, T_{\text{culture,pair}})$.
- $T_{\text{S3},i}$: Operationally assigned Sepsis-3 onset timestamp.

When blood culture precedes antimicrobial prescription ($T_{\text{culture,pair}} < T_{\text{abx,pair}}$), $T_{\text{susp}} = T_{\text{culture,pair}} \neq T_{\text{abx,pair}}$. Let $h = 1$ hour and define the dimensionless lookback coordinate $x_i = (T_{\text{susp},i} - 24\text{h} - T_{\text{admit},i})/h$. The search window evaluates

discrete hourly bins relative to admission, starting at integer index $\texttt{baseline_start_h}_i = \lfloor x_i \rfloor$ and ending at $\texttt{window_end_h}_i = \lfloor (T_{\text{susp},i} + 12\text{h} - T_{\text{admit},i})/h \rfloor$. When $\text{frac}(x_i) > 0$, discrete flooring places $\lfloor x_i \rfloor$ up to < 1h earlier than the continuous boundary $T_{\text{susp}} - 24$h. Because the upper boundary bin is floored, evaluated hours never exceed $T_{\text{susp}} + 12$h, guaranteeing $T_{\text{susp},i} - T_{\text{S3},i} \geq -12.0$h (empirical minimum: -11.2h). For episodes where the qualifying SOFA difference $\text{SOFA}(T_{\text{S3},i}) - \text{SOFA}_{\text{base},i} \geq 2$ is first identified at the earliest search bin ($N = 274/408$, 67.2% of all incident sepsis; and 100/151, 66.2% of surveillance-covered stays)—occurring when organ dysfunction relative to the lookback minimum $\text{SOFA}_{\text{base}}$ is already established at the earliest evaluated bin (where baseline minimum across $[T_{\text{susp}} - 24\text{h}, T_{\text{susp}}]$ may occur after the initial bin if dysfunction subsequently improves)—onset is assigned as $T_{\text{S3},i} = T_{\text{admit},i} + h\lfloor x_i \rfloor$, yielding the operational lead gap:

$$G_i = T_{\text{susp},i} - T_{\text{S3},i} = 24\text{h} + h\ \text{frac}(x_i) \tag{1}$$

Because $0 \leq \text{frac}(x_i) < 1$ (dimensionless), first-bin-assigned episodes satisfy $24\text{h} \leq G_i < 25\text{h}$, with an analytical supremum of 25h (the reported 24.99h is the observed empirical maximum). In these first-bin episodes (100/151, 66.2% of surveillance-covered stays), pre-suspicion opportunity ($t < T_{\text{susp}}$) is algebraically guaranteed for any decision time evaluated prior to onset ($t < T_{\text{S3}} \approx T_{\text{susp}} - 24\text{h}$), and the 2-hour buffer is trivially satisfied (100/100 eligible under both $\delta = 0$h and $\delta = 2$h). For the remaining 51 non-first-bin covered stays (33.8%), the median lead gap was 19.3h (IQR: 7.1–22.0h; 45 of 51 exhibiting $T_{\text{susp}} > T_{\text{S3}}$). Because evaluated early-warning decisions occur up to 6h prior to onset ($t < T_{\text{S3}}$), pre-suspicion opportunity holds for 50 of the 51 non-first-bin stays under $\delta = 0$h (1 ineligible stay, detailed in Supplementary Audit Table) and 49 of 51 under $\delta = 2$h (2 ineligible stays). Across all 151 covered stays, pre-suspicion eligibility is thus $100 + 50 = 150$, corresponding to 150/151 (99.3%) under $\delta = 0$h and $100 + 49 = 149$, corresponding to 149/151 (98.7%) under $\delta = 2$h. Thus, high pre-suspicion opportunity largely reflects this operational retrospective lookback convention rather than independent empirical evidence of unprompted biological warning. Across the full 408 cohort, the median lead gap was +24.2h (IQR: 22.8h to 24.6h; range: -11.2h to +24.99h).

2.3 Operational Cascade and Outcome-Stratified Hybrid Risk Sets

Let C denote the evaluated test cohort of unique ICU stays ($N = 8,254$), and let $\mathcal{T}_i \subseteq \{19, 20, \ldots, 30\}$ denote the set of hourly decision bins evaluated for stay i. For patients who never develop operational Sepsis-3 during hospitalization, we set $T_{\text{S3},i} = \infty$.

The full evaluated early-warning risk set is defined as:

$$\mathcal{R} = \left\{(i,t) : i \in C,\ t \in \mathcal{T}_i,\ t < T_{\text{S3},i}\right\} \tag{2}$$

The binary early-warning prediction target at anticipation horizon $H = 6$h is:

$$Y_i(t;H) = \mathbf{1}\left\{t < T_{\text{S3},i} \leq t + H\right\} \tag{3}$$

In the canonical unconstrained risk set ($\mathcal{R}_{\text{canonical}} = \mathcal{R}$), this definition yields exactly 97,700 evaluated hourly windows, comprising 678 horizon-positive observations ($Y = 1$) and 97,022 negative observations ($Y = 0$, prevalence $\pi = 0.694\%$).

The four stages of the operational cascade ($S \to V \to O \to D$) are formalized via stay-level indicators: $S_i = \mathbf{1}\{T_{\text{S3},i} < \infty\}$, $V_i = \mathbf{1}\{\exists t \in \mathcal{T}_i : Y_i(t;6\text{h}) = 1\}$, $O_i(\delta) = \mathbf{1}\{\exists t \in \mathcal{T}_i : Y_i(t;6\text{h}) = 1 \wedge t + \delta \leq C_i\}$, and $D_i(q,\delta) = \mathbf{1}\{i \in \mathcal{S}_{\text{alerted}}(K) \wedge Y_i(t_{\text{first},i};6\text{h}) = 1 \wedge t_{\text{first},i} + \delta \leq C_i\}$. To evaluate restricted decision windows, we define the outcome-stratified hybrid risk sets relative to reference clock C_i:

$$\mathcal{R}_\delta^{\text{hybrid}} = \left\{(i,t) \in \mathcal{R} : t + \delta \leq C_i\right\} \tag{4}$$

where $C_i = T_{\text{susp},i} = \min(T_{\text{culture,pair},i}, T_{\text{abx,pair},i})$ for sepsis stays ($S_i = 1$; determined by $T_{\text{abx,pair}}$ in 399 stays and culture charttime in 9 stays), and $C_i = T_{\text{Rx,start},i}$ for controls ($S_i = 0$; $C_i = \infty$ if unprescribed). For controls, 2,287 stays had $T_{\text{Rx,start}} \leq 19$h (censoring 27,388 windows); among 83 later-prescription controls, 21 censored 113 windows during surveillance ($65,926 = 93,427 - 27,388 - 113$), and 49 additional windows were removed at $\delta = 2$h ($65,877$). Combined with 4,266 sepsis windows (674 positive, 3,592 negative; 7 censored vs. canonical), $\mathcal{R}_0^{\text{hybrid}}$ evaluates 70,192 windows across 5,967 stays (prevalence 0.960%). Moving to $\delta = 2$h removes 63 windows ($70,129$ remaining: 65,877 control; 4,252 case windows, comprising 663 positive [$674 \to 663$] and 3,589 negative [$3,592 \to 3,589$]; prevalence 0.945%).

Importantly, cases and controls used different anchors; differences from canonical discrimination reflect both evaluated subsets and outcome-dependent selection, not a uniform policy. Under declared midnight assignment, all 408 sepsis stays had documented prescription start dates prior to hour 3.83 of ICU admission ($T_{\text{Rx,start}} \leq 3.83$h, median -12.6h) because every patient received empiric or prophylactic antimicrobial orders on or before admission day, whereas qualifying acute suspicion pairs ($T_{\text{abx,pair}}$) occurred later (median +50.7h). Consequently, all 4,273 case windows were excluded from earliest-prescription sets as post-prescription ($t \geq 19\text{h} > T_{\text{Rx,start}}$: 678 positive, 511 baseline negative, and 3,084 late-onset negative windows). Surviving observations in $\mathcal{R}_0^{\text{Rx}}$ (5,559 stays, 65,926 negative windows, 0 positive) and $\mathcal{R}_2^{\text{Rx}}$ (65,877 negative windows; $N_+^{\text{Rx}} = 0$; prevalence strictly 0.000%) derive solely from controls. Discrimination was thus mathematically unestimable: AUROC is undefined and AUPRC uninformative.

2.4 Alert-Budget Operating Frontiers and Refractory Policy

To quantify simulated emitted-alert burden, we evaluated fixed alert budgets $q \in \{5.0, 10.0, 20.0\}$ emitted alert episodes per 100 surveillance patient-days, targeting $K = \text{round}(q \cdot E/100)$ alert episodes across total surveillance exposure $E = \sum_i h_i/24 = 4,070.8$ patient-days (yielding $K = 204, 407, 814$). Because surveillance spans an 11-hour interval ($t \in \{19, \ldots, 30\}$), a 12-hour refractory suppression rule guarantees at most one alert episode per patient stay ($N_{\text{episodes},i} \leq 1$), as any alert emitted at $t \in [19, 30]$ silences subsequent alerts past hour 30. Simulating chronological refractory alerts is thus algebraically equivalent to selecting the top K patient stays ranked by their peak surveillance score $\max_{t \in \mathcal{T}_i} s_i(t)$. We report three operational metrics: Episode PPV $= N_{\text{det}}/K$, Covered Sensitivity $= N_{\text{det}}/151$, and Population Yield $= N_{\text{det}}/408$, where timely detections $N_{\text{det}}(q,\delta)$ are defined as:

$$N_{\text{det}}(q,\delta) = \sum_{i \in \mathcal{S}_{\text{alerted}}(K)} \mathbf{1}\{S_i = 1 \wedge Y_i(t_{\text{first},i};6\text{h}) = 1 \wedge t_{\text{first},i} + \delta \leq C_i\} \tag{5}$$

Alerts were selected across the full canonical stream $\mathcal{R}$. Selecting patients by peak score $\max_t s_i(t)$ establishes batch threshold-solving equivalence, with timing adjudicated at first threshold crossing: $t_{\text{first},i} = \min\{t \in \mathcal{T}_i : s_i(t) \geq \tau(K)\}$, under deterministic tie-breaking $(-\max_t s_i(t), \text{ICUSTAY_ID})$. Because surveillance-covered sepsis cases are censored at onset (mean 7.87 evaluated bins, vs 10.47 across all 408 sepsis stays) while controls are observed across a mean of 11.91 bins (12 bins for uncensored controls; 180 controls lose 725 windows through observation cessation),

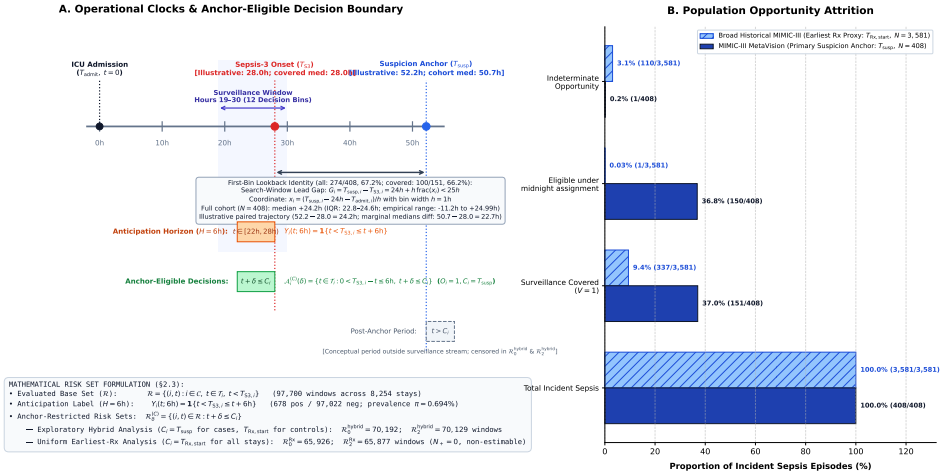


Figure 1: Operational Clocks and Anchor-Dependent Evaluation Eligibility in the MIMIC-III MetaVision Cohort. The timeline is schematic; displayed population summaries do not represent a single paired patient trajectory. Eligibility relative to a retrospective suspicion anchor does not establish absence of prior antimicrobial exposure. Panel A depicts the clinical timeline for an ICU-incident sepsis stay, illustrating the discrete search-window lead $G_i = 24\text{h} + h\ \mathrm{frac}(x_i) < 25\text{h}$ for first-bin-assigned episodes ($N = 274/408$, 67.2%; median +24.2h; empirical max 24.99h), the anticipation horizon $t \in [22\text{h}, 28\text{h})$, anchor-eligible decision times $\mathcal{A}_i^{(C)}(\delta)$, and the conceptual post-anchor period. Actual model evaluations occur at discrete scored timestamps $t \in \mathcal{T}_i$. Panel B displays population opportunity attrition comparing Broad Historical MIMIC-III (earliest prescription proxy $T_{\text{Rx,start}}$, $N = 3,581$) and MIMIC-III MetaVision (suspicion anchor T_{susp}, $N = 408$), reporting exact stay counts beside each opportunity tier under declared midnight assignment; this comparison reflects concurrent differences in both cohort selection and anchor definition rather than isolating an anchor effect alone. *Alt-text:* Two-panel diagram displaying schematic timeline clocks and formal risk-set definitions in Panel A, and horizontal bar charts comparing surveillance coverage and treatment-precedence opportunity in Panel B.

Table 1: Cohort Progression and Denominator Accounting Ledger. Denominators traced from the base test cohort through surveillance eligibility, sepsis onset timing, prediction horizons, and pre-treatment actionability. *Note:* Rows 1–3b report cohort shares; rows 4–6 report window-level positive prevalence (π). Historical cohort ($N = 3,581$) comprises CareVue-only (1,687), MetaVision-only (1,886), and 8 transitional stays ($1,687 + 1,886 + 8 = 3,581$).

Progression Stage / Cohort Partition	Stays (N)	Evaluated Windows	Surveillance Volume	Prev. / Share	Clinical Definition & Measurement Convention
1. Base Prospective-Replay Test Cohort	8,254	97,700 / 99,048 sched.	4,070.8 pt-days	100.0%	Adult first ICU stays $\geq$ 30h; 1,348 censored windows (623 onset, 725 cessation)
2. Non-Sepsis Control ICU Stays	7,846	93,427 neg. windows	3,892.8 pt-days	95.06%	Stays without operational Sepsis-3 during hospitalization
3. Operational Sepsis-3 Stays ($S = 1$)	408	4,273 (678 pos, 3,595 neg)	178.0 pt-days	4.94%	Blood culture + antibiotic + acute ΔSOFA ≥ 2; full cohort median onset 49.0h
3a. Late-Onset Sepsis (Onset > 36h)	257	3,084 neg. windows	128.5 pt-days	63.0% of septic	Onset > 36h (median: 69.0h); anticipation horizon $[T_{S3} - 6, T_{S3}]$ outside [19, 30] ($V = 0$)
3b. Surveillance-Covered ($V = 1$)	151	1,189 (678 pos, 511 neg)	49.5 pt-days	37.0% of septic	Onset in [20h, 36h] (median: 28.0h); anticipation horizon intersects window ($V = 1$, earliest uncensored integer onset bin is 20h)
4. Canonical Sepsis-3 Risk Set ($H = 6$h)	8,254	97,700 (678 pos / 97,022 neg)	4,070.8 pt-days	0.694%	Full unconstrained risk set ($t \in \mathcal{T}_i$, $t < T_{S3,i}$)
5. Outcome-Stratified Hybrid Set $\mathcal{R}_0^{\text{hybrid}}$ ($\delta = 0$h)	5,967	70,192 (674 pos / 69,518 neg)	2,924.7 pt-days	0.960%	Censoring post-anchor ($t \leq C_i$; T_{susp} cases, $T_{\text{Rx,start}}$ controls); 150/151 covered eligible
6. Outcome-Stratified Hybrid Set $\mathcal{R}_2^{\text{hybrid}}$ ($\delta = 2$h)	5,967	70,129 (663 pos / 69,466 neg)	2,922.0 pt-days	0.945%	2-hour proxy-timing margin ($t + 2\text{h} \leq C_i$); 149/151 covered eligible

differential stay durations introduce multiple-testing duration bias under peak-score aggregation, as characterized by Do et al. [19].

2.5 Predictive Models and Statistical Inference

We evaluated 8 models across three paradigms: (1) State-Space Models (BiMamba SSM) [25, 26]; (2) Deep Sequence Transformers (Mask-Aware Clinical Transformer and Baseline Unmasked Transformer); (3) Tabular and Ensemble Baselines (Random Forest, XGBoost, LightGBM, L2 Logistic Regression, and Multi-Model Stacking Ensemble). The 21 features comprised 6 vitals (heart rate, systolic, diastolic, and mean arterial pressure, respiratory rate, SpO_2) and 15 laboratory/clinical variables (WBC, platelets, creatinine, BUN, total bilirubin, lactate, glucose, sodium, potassium, bicarbonate, hematocrit, PaO_2, FiO_2, urine output, GCS). Missing values were forward-filled (up to 4h for vitals, 24h for labs) and median-imputed. Following established 20-hour lookback and sequential feature engineering formulations in ICU deep learning [27], lookback $[t - 20\text{h}, t)$ strictly precedes decision time t (pre-admission bin $[-1\text{h}, 0\text{h})$ is evaluated only at $t = 19$h and zero-masked in sequence models). Tabular models received the flattened $20 \times 21 = 420$-dimensional vector; sequence models processed length-$L = 20$ normalized series (42 channels for Mask-Aware Transformer using binary masks $\mathbf{m}_{i,t} \in \{0, 1\}^{21}$). Deep models used weighted binary cross-entropy with hidden dimension 64 across 2 layers (AdamW); tree ensembles used 300 estimators (depth 6, learning rate 0.05). The Stacking Ensemble combined out-of-fold predictions from five base learners via an L2 logistic meta-learner. BiMamba SSM mean-pooled hidden states with linear $O(L)$ scan complexity. We excluded discrete integer heuristics (qSOFA, NEWS2) whose step-function distributions with pervasive ties resist continuous alert-budget tuning, restricting our scope to continuous multivariate ML models.

Uncertainty was quantified using patient-clustered paired bootstrapping ($B = 1,000$ resamples). For Procedure B (Empirical Burden-Matched Bootstrap), exposure $E_b = \frac{1}{24}\sum_{i \in b} h_i$, integer alert quota $K_b = \mathrm{round}(q \cdot E_b/100)$, and resample-specific denominators ($N_{V,b} = \sum_{i \in b} V_i$ for covered sensitivity; $N_{S,b} = \sum_{i \in b} S_i$ for population yield) were dynamically recomputed in each resample. NNE confidence intervals were derived by exact reciprocal transformation from unrounded PPV_b: $\text{NNE}_{\text{low}} = 100/\text{PPV}_{\text{upper}}$ and $\text{NNE}_{\text{high}} = 100/\text{PPV}_{\text{low}}$.

3 Results

3.1 Cohort Progression and Denominator Accounting

Table 1 traces all 8,254 ICU admissions through surveillance eligibility, onset timing, lookback horizons, and pre-treatment actionability. Across all 408 operational Sepsis-3 stays, median

onset was 49.0h post-admission. Exactly 257 stays (63.0%; median onset: 69.0h) had clinical onset after hour 36. Under the specified observation schedule and 6-hour horizon, these 257 stays had no evaluated horizon-positive decision time ($V = 0$). The remaining 151 stays (37.0%; median onset: 28.0h) had active protocol surveillance coverage ($V = 1$). Of the 151 covered stays, 150 had at least one horizon-positive decision time at or before the selected infection-suspicion anchor (99.3% within-covered opportunity; 36.8% of total incident sepsis under $\delta = 0$h; and 149 stays, 98.7% under $\delta = 2$h).

3.2 Outcome-Stratified Hybrid Discrimination Analysis

Table 2 presents discrimination across Canonical Sepsis-3, Hybrid Set ($\mathcal{R}_0^{\text{hybrid}}$), and Hybrid Set with 2h Buffer ($\mathcal{R}_2^{\text{hybrid}}$). Discrimination changed little in the outcome-stratified hybrid analysis (BiMamba SSM: AUROC 0.6326 canonical vs. 0.6433 on $\mathcal{R}_0^{\text{hybrid}}$, paired ΔAUROC = +0.0107, 95% CI: +0.0060 to +0.0175; Stacking Ensemble: 0.6284 vs. 0.6351, paired ΔAUROC = +0.0067). Because cases and controls were restricted using different anchors, this finding does not establish preserved discrimination under a uniform pre-suspicion or pre-prescription policy. The increase in raw AUPRC (BiMamba: 1.209% to 1.656%) accompanied control-window censoring and higher positive prevalence (raised by 1.384×, from $\pi = 0.694\%$ to 0.960%); these comparisons do not isolate improved ranking performance. When evaluated as prevalence-normalized lift (AUPRC/π), discriminative gain over baseline remains essentially flat (1.742× canonical vs. 1.725× on $\mathcal{R}_0^{\text{hybrid}}$ for BiMamba, and 1.738× vs. 1.688× for the Stacking Ensemble), consistent with a substantial contribution from prevalence and evaluated-subset changes; these comparisons do not isolate changes in ranking performance. These results are directionally consistent with the order-based sensitivity findings of Dutta et al. [18].

3.3 Observed Burden–Detection Trade-Off Under This Protocol

Table 3 and Figure 2 demonstrate the observed trade-off under the prespecified alert-budget scenarios: bedside detection yield is constrained by alert-rate limits. Alerts were selected across the full canonical surveillance stream (4,070.8 patient-days; 97,700 windows across all 8,254 test stays); C_i was used only retrospectively to adjudicate advance-warning timing ($t_{\text{first},i} + \delta \le T_{\text{susp},i}$ for sepsis cases), rather than excluding observations prior to alert selection. Retaining control stays with prior antimicrobial orders in the alert pool simulates an unconstrained clinical screening deployment; in live practice, suppressing alerts on patients already receiving broad-spectrum antimicrobials would be an operational deployment option that prevents treated controls from consuming alert capacity.

At an emitted workload of 10 alert episodes per 100 patient-days ($K = 407$ emitted episodes across 4,070.8 patient-days), episode-level PPV across all 8 models ranged from 0.98% to 1.72%, corresponding to 58.1 to 101.8 simulated alert evaluations per detected patient (NNE). BiMamba SSM and LightGBM achieved the highest PPV at 1.72% (95% CI: 0.49–2.95%; 7 detected cases out of 150 eligible; NNE = 58.1, 95% CI: 33.9–203.5; covered sensitivity 4.64%; population yield 1.72%). Stacking Ensemble, XGBoost, and Random Forest detected 6 cases each (PPV 1.47%, NNE = 67.8); Mask-Aware Transformer and Clinical Transformer detected 5 cases each (PPV 1.23%, NNE = 81.4); and L2-Logistic Regression detected 4 cases (PPV 0.98%, NNE = 101.8). At 20 episodes/100 pt-days, XGBoost detected 15 patients (NNE = 54.3) and Mask-Aware Transformer detected 14 patients (NNE = 58.1). In this sample, horizon-correct alerts and alerts satisfying the two-hour pre-suspicion margin ($t_{\text{first},i} + 2\text{h} \le T_{\text{susp},i}$) were identical across all models (7 of 7 for BiMamba SSM), occurring before suspicion (T_{susp}) rather than earliest prescribing ($T_{\text{Rx,start}}$).

Of the 407 emitted alerts for BiMamba SSM, 54 fired on sepsis stays (7 timely within $(0, 6]$h; 47 premature > 6h prior to onset) and 353 on control stays. Crucially, 71 control alerts (17.4% of all alerts; 20.1% of control alerts) fired on pre-treated controls ($T_{\text{Rx,start}} \le 19$h), while 277 fired on untreated controls, and 5 on controls with prescriptions after hour 30. Across models, pre-treated controls consumed 17.4% (BiMamba SSM, Stacking Ensemble; 71/407) to 31.4% (LightGBM; 128/407) of all alerts. With 7 timely detections ($N_{\text{false}} = 407 - 7 = 400$), retrospectively removing alerts on pre-treated controls removes 17.8% (71/400) and 32.0% (128/400) of false alerts for BiMamba SSM and LightGBM, respectively. These counts describe retrospective alert occupancy; a prospective suppression rule would apply to all patients and require verified active administration rather than date-coarsened orders, which would otherwise silence all 408 sepsis stays. Real-time effects on volume, detection, and safety were not prospectively evaluated.

3.4 Two-Axis Subgroup and Timestamp Uncertainty Matrix

Table 4 presents the cross-tabulation of Sepsis Onset Timing against Treatment Proxy Timing. Because MIMIC-III prescription start dates (`prescriptions.startdate`) lack time of day, order initiation is evaluated both under declared point assignment (midnight a_i) and under calendar-day intervals $[a_i, b_i) = [\text{midnight}, \text{next midnight})$. For stay i, let minute-resolved culture charttime be c_i, coarsened prescription interval be $[a_i, b_i)$, and pre-onset evaluation windows be $P_i = \{t \in \mathcal{T}_i : 0 < T_{\text{S3},i} - t \le H\}$. Under the infection-suspicion anchor $T_{\text{susp},i} = \min(A_i, c_i)$ with unobserved $A_i \in [a_i, b_i)$, the attainable set C_i of $T_{\text{susp},i}$ is $\{c_i\}$ if $c_i < a_i$, $[a_i, c_i]$ if $a_i \le c_i < b_i$, and $[a_i, b_i)$ if $c_i \ge b_i$. Because $C_i = c_i$ is attained whenever $A_i \ge c_i$, the upper endpoint is closed when $a_i \le c_i < b_i$. Definite opportunity requires $t + \delta \le \min(A_i, c_i)$ for all $A_i \in [a_i, b_i)$, holding if and only if $t + \delta \le a_i$ and $t + \delta \le c_i$:

$$O_i^{\text{def}}(\delta) = \mathbf{1}\{\exists t \in P_i : t + \delta \le a_i \ \wedge \ t + \delta \le c_i\} \tag{6}$$

Possible opportunity requires $t + \delta \le \min(A_i, c_i)$ for at least one $A_i \in [a_i, b_i)$, which holds if and only if $t + \delta < b_i$ and $t + \delta \le c_i$:

$$O_i^{\text{poss}}(\delta) = \mathbf{1}\{\exists t \in P_i : t + \delta < b_i \ \wedge \ t + \delta \le c_i\} \tag{7}$$

For the earliest prescription proxy alone ($T_{\text{Rx,start}}$):

$$\begin{aligned} O_i^{\text{def}}(\delta) &= \mathbf{1}\{\exists t \in P_i : t + \delta \le a_i\}, \\ O_i^{\text{poss}}(\delta) &= \mathbf{1}\{\exists t \in P_i : t + \delta < b_i\} \end{aligned} \tag{8}$$

Stays with $O_i^{\text{def}} = 0$ but $O_i^{\text{poss}} = 1$ are indeterminate ($O_i^{\text{indet}} = 1$). All indicators strictly satisfy $O_i^{\text{def}} \le O_i^{\text{point}} \le O_i^{\text{poss}}$ and $O_i^{\text{poss}} = O_i^{\text{def}} + O_i^{\text{indet}}$, with $O_i^{\text{def}} \equiv O_i^{\text{point}}$ by construction because midnight point assignment $a_i = \inf[a_i, b_i)$. Importantly, these opportunity bounds are strictly conditional on fixed operational Sepsis-3 timestamps and surveillance coverage; they do not propagate counterfactual antibiotic timings through the upstream phenotype (which could alter qualifying pairs, baseline SOFA, onset, or coverage).

Opportunity depended strongly on the selected clinical-process anchor. Under the infection-suspicion anchor (T_{susp}), 150 of 151

Table 2: Benchmark Discrimination Across Outcome-Stratified Hybrid Risk Sets (Held-Out Point Estimates and Clustered 95% CIs). Point estimates calculated directly on held-out test data ($B = 0$); 95% CIs derived from $B = 1,000$ patient-clustered bootstrap resamples. Evaluated on hybrid risk sets $\mathcal{R}_0^{\text{hybrid}}$ and $\mathcal{R}_2^{\text{hybrid}}$ censoring cases at infection-suspicion anchor T_{susp} and controls at earliest prescription $T_{\text{Rx,start}}$. Sepsis cases and controls were restricted using different clinical-process anchors; differences from canonical discrimination reflect both evaluated observations and outcome-specific selection, and do not represent a uniform prospective eligibility policy.

Model Architecture / Metric	Canonical Sepsis-3 (H = 6h)	Hybrid $\mathcal{R}_0^{\text{hybrid}}$ (δ = 0h)	Hybrid $\mathcal{R}_2^{\text{hybrid}}$ (δ = 2h)	Paired ΔAUROC ($\mathcal{R}_0^{\text{hybrid}}$ − Canon)
BiMamba SSM	0.6326 (0.5872–0.6733)	**0.6433** (0.5998–0.6835)	**0.6447** (0.6002–0.6860)	**+0.0107** (+0.0060 to +0.0175)
AUPRC	*1.209% (0.892–1.769)*	***1.656%** (1.235–2.329)*	***1.647%** (1.220–2.331)*	—
Multi-Model Stacking Ensemble	0.6284 (0.5829–0.6699)	0.6351 (0.5897–0.6771)	0.6370 (0.5913–0.6799)	+0.0067 (+0.0021 to +0.0137)
AUPRC	*1.206% (0.882–1.748)*	*1.621% (1.207–2.263)*	*1.613% (1.198–2.255)*	—
Random Forest	0.6108 (0.5643–0.6535)	0.6143 (0.5684–0.6575)	0.6173 (0.5712–0.6612)	+0.0034 (-0.0008 to +0.0101)
AUPRC	*1.080% (0.821–1.434)*	*1.470% (1.138–1.920)*	*1.461% (1.120–1.916)*	—
Mask-Aware Clinical Transformer	0.5942 (0.5467–0.6410)	0.5961 (0.5490–0.6422)	0.5954 (0.5480–0.6426)	+0.0020 (-0.0024 to +0.0083)
AUPRC	*1.013% (0.766–1.303)*	*1.382% (1.048–1.760)*	*1.365% (1.032–1.752)*	—
Tabular GBDT (XGBoost)	0.5670 (0.5208–0.6102)	0.5643 (0.5186–0.6082)	0.5652 (0.5192–0.6099)	-0.0027 (-0.0070 to +0.0038)
AUPRC	*0.977% (0.739–1.357)*	*1.320% (1.008–1.817)*	*1.313% (0.998–1.811)*	—
Tabular GBDT (LightGBM)	0.5613 (0.5161–0.6026)	0.5587 (0.5139–0.6010)	0.5598 (0.5145–0.6033)	-0.0026 (-0.0070 to +0.0038)
AUPRC	*0.893% (0.694–1.195)*	*1.222% (0.957–1.629)*	*1.214% (0.949–1.630)*	—
Logistic Regression (L2)	0.5593 (0.5097–0.6048)	0.5654 (0.5157–0.6107)	0.5657 (0.5150–0.6127)	+0.0061 (+0.0023 to +0.0116)
AUPRC	*0.939% (0.716–1.218)*	*1.337% (1.012–1.740)*	*1.325% (1.002–1.731)*	—
Clinical Transformer (Baseline)	0.5239 (0.4808–0.5653)	0.5296 (0.4851–0.5712)	0.5286 (0.4834–0.5705)	+0.0057 (+0.0019 to +0.0110)
AUPRC	*0.808% (0.609–1.381)*	*1.147% (0.857–1.904)*	*1.131% (0.845–1.894)*	—

*Paired differences (ΔAUROC) were calculated from unrounded model predictions prior to display rounding.

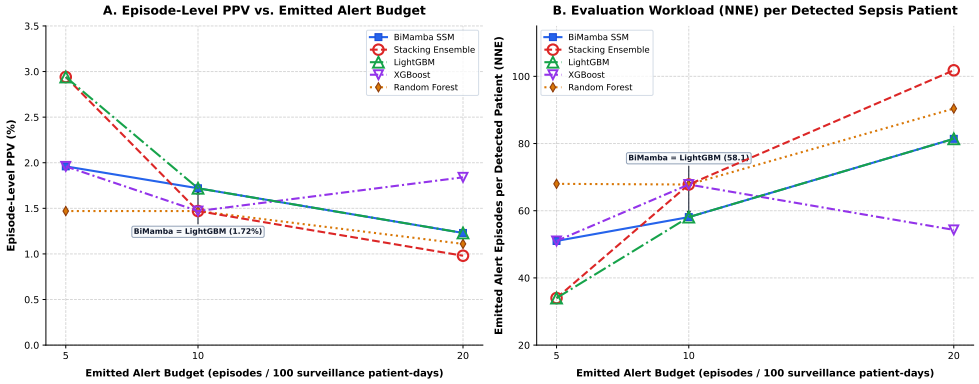


Figure 2: Operational Alert-Budget Performance Frontier. Panel A displays episode-level positive predictive value (PPV %) across prespecified alert budgets. Panel B displays the Number Needed to Evaluate (NNE) per detected patient under a 12-hour alert suppression rule. Lines connect discrete prespecified operating budgets (5, 10, and 20 emitted episodes per 100 patient-days) for a representative 5-model subset; complete 8-model results, bootstrap confidence intervals, and boundary-degenerate notes are reported in Table 3. BiMamba SSM and LightGBM have coincident operating points at budgets of 10 and 20 episodes per 100 surveillance patient-days. *Alt-text:* Two-panel plot showing PPV (Panel A) and evaluation workload per detected patient (Panel B) across alert budgets of 5, 10, and 20 episodes/100 patient-days for 5 models under 12h alert suppression.

covered stays (99.3% within-covered; 36.8% of total incident sepsis) had decision bins satisfying the definite opportunity condition under declared midnight point assignment a_i. Across the calendar-day interval, exactly 150 stays satisfy $O_i^{\text{def}} = 1$, with only 1 indeterminate stay ($O_i^{\text{indet}} = 1$), bounding opportunity between 36.8% and 37.0% ($[150, 151]/408$; $[99.3\%, 100.0\%]$ within-covered). In contrast, under the earliest hospital prescription-start proxy ($T_{\text{Rx,start}}$ where $L_i = a_i, U_i = b_i$), exactly 0 covered stays had definite opportunity under midnight point-assignment ($O_i^{\text{def}} = 0, O_i^{\text{point}} = 0$), with 47 indeterminate stays ($O_i^{\text{indet}} = 47$), establishing bounds $[0.0\%, 11.5\%]$ ($[0, 47]/408$; $[0.0\%, 31.1\%]$ within-covered). Because no horizon-positive cases remained before point-assigned earliest prescriptions, this task was not evaluable for discrimination under midnight point assignment. In broad historical MIMIC-III ($N = 3,581$), intervals generate 1 definitely eligible stay, 110 indeterminate stays, and 111 possibly eligible stays ($[1, 111]/337 = [0.30\%, 32.94\%]$ within-covered; whole-cohort bounds $[0.03\%, 3.10\%]$ $[[1, 111]/3,581]$). The 150/151 result under T_{susp} does not establish that patients were unexposed to antimicrobials or that models predicted prior to first administration. Equally, zero definite opportunity under $T_{\text{Rx,start}}$ reflects date coarsening, not verified bedside exposure.

3.5 Cumulative Population Coverage Decomposition

Figure 3 illustrates the four-stage cumulative population coverage decomposition ($S \to V \to O \to D$):

$$\frac{|V=1|}{|S=1|} \times \frac{|O=1|}{|V=1|} \times \frac{|D=1|}{|O=1|} = \frac{151}{408} \times \frac{150}{151} \times \frac{7}{150} = \frac{7}{408} = 1.72\%$$

1. **Surveillance Truncation** ($151/408 = 37.0\%$)**:** 63.0% of sepsis patients develop onset after hour 36, outside the prespecified second-day surveillance protocol.
2. **Suspicion-Anchor Eligibility** ($150/151 = 99.3\%$)**:** Of the 151 covered stays, 150 had at least one horizon-positive decision time at or before the retrospective suspicion anchor (T_{susp}) [149 of 151 (98.7%) under a 2-hour pre-suspicion margin]. Together, surveillance coverage and suspicion-anchor eligibility impose a theoretical population ceiling of $150/408 = 36.76\%$.
3. **Model-Alert Policy Detection** ($7/150 = 4.67\%$)**:** At 10 alert episodes/100 patient-days ($K = 407$), an oracle ranking policy could theoretically detect all 150 eligible patients (PPV $150/407 = 36.86\%$). Under the empirical score ranking of BiMamba SSM with 12-hour alert suppression, the model detects 7 distinct patients (4.67% of anchor-eligible stays; 4.64% of covered stays [7/151]; 1.72% whole-population yield [7/408]; NNE = 58.1). The low observed yield reflects performance of the tested score-and-first-alert policies within the restricted surveillance protocol, rather than a budget-imposed ceiling near 2%.

Table 3: Operational Burden-Matched Alert Frontiers Under 12h Refractory Suppression. Evaluated under Procedure B with exposure recomputation and first-emitted-alert timing verification.

Budget	Model Architecture	Cutoff (τ)	Alerts (K)	Episode PPV % [95% CI]	Det (N)	Sens. Cov %	Yield %	NNE [95% CI]
5.0 ep / 100 pt-d	Stacking Ensemble	0.0145	204	2.94 (0.98–5.88)%	6	3.97 (1.31–7.65)%	1.47 (0.49–2.86)%	34.0 (17.0–102.0)
	LightGBM	0.0117	204	2.94 (0.98–5.39)%	6	3.97 (1.26–7.10)%	1.47 (0.47–2.61)%	34.0 (18.5–102.0)
	BiMamba SSM	0.0187	204	1.96 (0.49–4.45)%	4	2.65 (0.65–6.20)%	0.98 (0.24–2.31)%	51.0 (22.5–204.0)
	XGBoost	0.0122	204	1.96 (0.49–3.94)%	4	2.65 (0.63–5.63)%	0.98 (0.23–2.09)%	51.0 (25.4–204.0)
	Random Forest	0.0177	204	1.47 (0.00–3.45)%	3	1.99 (0.00–4.49)%	0.74 (0.00–1.70)%	68.0 (29.0–∞)
	Clinical Transformer	0.0101	204	0.98 (0.00–2.46)%	2	1.32 (0.00–3.50)%	0.49 (0.00–1.27)%	102.0 (40.6–∞)
	Logistic Regression	0.0103	204	0.98 (0.00–2.94)%	2	1.32 (0.00–3.85)%	0.49 (0.00–1.45)%	102.0 (34.0–∞)
	Mask-Aware Transformer	0.0120	204	0.00 (0.00–0.00)%*	0	0.00 (0.00–0.00)%*	0.00 (0.00–0.00)%*	Undefined (∞)
10.0 ep / 100 pt-d	BiMamba SSM	0.0165	407	1.72 (0.49–2.95)%	7	4.64 (1.33–8.16)%	1.72 (0.49–2.93)%	58.1 (33.9–203.5)
	LightGBM	0.0110	407	1.72 (0.49–2.95)%	7	4.64 (1.38–7.74)%	1.72 (0.49–2.89)%	58.1 (33.9–203.5)
	Stacking Ensemble	0.0131	407	1.47 (0.25–2.46)%	6	3.97 (0.70–6.98)%	1.47 (0.26–2.54)%	67.8 (40.7–407.0)
	XGBoost	0.0114	407	1.47 (0.49–2.94)%	6	3.97 (1.32–7.65)%	1.47 (0.49–2.85)%	67.8 (34.0–203.5)
	Random Forest	0.0158	407	1.47 (0.49–2.70)%	6	3.97 (1.26–7.29)%	1.47 (0.47–2.70)%	67.8 (37.0–204.0)
	Mask-Aware Transformer	0.0110	407	1.23 (0.25–2.46)%	5	3.31 (0.74–6.67)%	1.23 (0.26–2.48)%	81.4 (40.7–407.0)
	Clinical Transformer	0.0094	407	1.23 (0.25–2.46)%	5	3.31 (0.68–6.38)%	1.23 (0.25–2.37)%	81.4 (40.7–407.0)
	Logistic Regression	0.0095	407	0.98 (0.25–1.97)%	4	2.65 (0.59–5.51)%	0.98 (0.23–2.07)%	101.8 (50.8–408.0)
20.0 ep / 100 pt-d	XGBoost	0.0105	814	1.84 (0.98–2.70)%	15	9.93 (5.26–14.47)%	3.68 (1.87–5.29)%	54.3 (37.0–101.9)
	Mask-Aware Transformer	0.0099	814	1.72 (0.74–2.58)%	14	9.27 (4.54–13.77)%	3.43 (1.59–5.19)%	58.1 (38.8–135.7)
	Logistic Regression	0.0088	814	1.35 (0.49–2.09)%	11	7.28 (2.74–11.11)%	2.70 (0.99–4.18)%	74.0 (47.8–203.7)
	BiMamba SSM	0.0142	814	1.23 (0.49–2.09)%	10	6.62 (3.07–10.89)%	2.45 (1.05–4.10)%	81.4 (47.9–203.2)
	LightGBM	0.0105	814	1.23 (0.61–2.09)%	10	6.62 (3.11–11.04)%	2.45 (1.15–4.10)%	81.4 (47.9–163.0)
	Random Forest	0.0131	814	1.11 (0.49–1.85)%	9	5.96 (2.70–9.68)%	2.21 (0.96–3.69)%	90.4 (54.2–203.7)
	Stacking Ensemble	0.0115	814	0.98 (0.37–1.72)%	8	5.30 (2.04–9.15)%	1.96 (0.74–3.43)%	101.8 (58.1–271.3)
	Clinical Transformer	0.0088	814	0.86 (0.25–1.48)%	7	4.64 (1.51–8.18)%	1.72 (0.52–3.04)%	116.3 (67.8–407.0)

*At zero observed detections, the empirical percentile-bootstrap interval was degenerate at zero. This does not establish zero population detection probability. NNE intervals were derived by exact reciprocal transformation from unrounded bootstrap $\text{PPV}_b = 100 \cdot N_{\text{det},b}/K_b$ under dynamically recomputed exposure E_b and budget $K_b = \text{round}(q \cdot E_b/100)$. Covered sensitivity and population yield denominators ($N_{V,b} = \sum_{i \in b} V_i$ and $N_{S,b} = \sum_{i \in b} S_i$) are recomputed dynamically within each resample, reflecting patient-cluster sampling variability. In this observed sample, horizon-correct detections and detections satisfying the two-hour pre-suspicion proxy-timing margin ($t_{\text{first},i} + 2\text{h} \le T_{\text{susp},i}$) were identical across all models and operating points.

Table 4: Two-Axis Subgroup and Timestamp Uncertainty Matrix. Opportunity bounds conditional on fixed operational Sepsis-3 timestamps and surveillance coverage. *Note:* Subgroup Bounds denote $[O^{\text{def}}, O^{\text{poss}}]/\text{Total}(N)$ within each onset stratum. For All Incident Sepsis-3, within-covered bounds ($[O^{\text{def}}, O^{\text{poss}}]/\text{Covered}[V = 1]$) are $[150, 151]/151 = [99.3\%, 100.0\%]$ for MetaVision T_{susp}, $[0, 47]/151 = [0.0\%, 31.1\%]$ for MetaVision Earliest Rx, and $[1, 111]/337 = [0.30\%, 32.94\%]$ for Broad Historical MIMIC-III.

Cohort & Timestamp Source	Sepsis Onset Category	Total (N)	Covered ($V = 1$)	Def. Elig (O^{def})	Point (O^{point})	Indet (O^{indet})	Poss (O^{poss})	Subgroup Bounds	Contribution to Cohort
MetaVision Prospective-Replay	Early Incident (≤ 24h)	47	47 (100.0%)	47	47	0	47	[100.0%, 100.0%]	[47, 47] / 408 (11.5%)
(Primary Benchmark: Suspicion Anchor T_{susp})	Intermediate (24–48h)	156	104 (66.7%)	103	103	1	104	[66.0%, 66.7%]	[103, 104] / 408 (25.2–25.5%)
	Late Incident (> 48h)	205	0 (0.0%)	0	0	0	0	[0.0%, 0.0%]	0 / 408 (0.0%)
	All Incident Sepsis-3	**408**	**151 (37.0%)**	**150**	**150**	**1**	**151**	**[36.8%, 37.0%]**	**[150, 151] / 408 (36.8–37.0%)**
MetaVision Sensitivity: Earliest Rx	Early Incident (≤ 24h)	47	47 (100.0%)	0	0	38	38	[0.0%, 80.9%]	[0, 38] / 408 (0.0–9.3%)
(Pre-Exposure Sensitivity: Earliest Rx $T_{\text{Rx,start}}$)	Intermediate (24–48h)	156	104 (66.7%)	0	0	9	9	[0.0%, 5.8%]	[0, 9] / 408 (0.0–2.2%)
	Late Incident (> 48h)	205	0 (0.0%)	0	0	0	0	[0.0%, 0.0%]	0 / 408 (0.0%)
	All Incident Sepsis-3	**408**	**151 (37.0%)**	**0**	**0**	**47**	**47**	**[0.0%, 11.5%]**	**[0, 47] / 408 (0.0–11.5%)**
Broad Historical MIMIC-III	Early Incident (≤ 24h)	2,641	102 (3.9%)	0	0	74	74	[0.00%, 2.80%]	[0, 74] / 3,581 (0.00–2.07%)
(Date-Level Proxy: Earliest Rx $T_{\text{Rx,start}}$)	Intermediate (24–48h)	361	235 (65.1%)	1	1	36	37	[0.28%, 10.25%]	[1, 37] / 3,581 (0.03–1.03%)
	Late Incident (> 48h)	579	0 (0.0%)	0	0	0	0	[0.00%, 0.00%]	[0, 0] / 3,581 (0.00%)
	All Incident Sepsis-3	**3,581**	**337 (9.4%)**	**1**	**1**	**110**	**111**	**[0.03%, 3.10%]**	**[1, 111] / 3,581 (0.03–3.10%)**

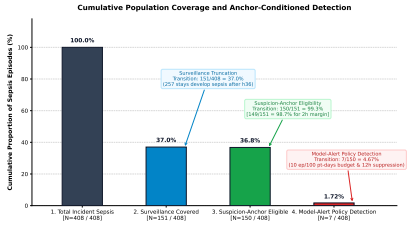


Figure 3: Cumulative Population Coverage and Anchor-Conditioned Detection. Stepwise attrition from 408 incident sepsis episodes through surveillance protocol coverage, retrospective suspicion-anchor eligibility, and model-alert policy detection. *Alt-text:* Four-stage bar chart showing patient opportunity attrition from total incident sepsis (408 patients) to surveillance coverage (151 patients, 37.0%), suspicion-anchor eligibility (150 patients, 36.8%), and timely model-alert policy detection (7 patients, 1.72%).

4 Discussion

This investigation characterizes the operational trade-offs governing machine learning early warning for sepsis in ICU surveillance:

1. **Outcome-Dependent Anchor Selection Governs Evaluability:** The canonical surveillance analysis quantified population coverage and emitted-alert yield. A secondary outcome-stratified hybrid restriction produced small changes in discrimination, but did not represent a uniform prospective eligibility policy. Under a common earliest-prescription-start proxy and declared midnight assignment, no eligible horizon-positive observations remained, preventing discrimination estimation for that restricted task.
2. **Detection Yield Is Constrained by Workload and Prevalence:** AUROC values near 0.64 are modest. At 10 emitted episodes per 100 evaluation-bin patient-days, PPV ranged from 0.98% to 1.72%, corresponding to 58.1–101.8 simulated evaluations per detected patient (NNE); population-level timely yield was 1.72% under the prespecified surveillance schedule.
3. **Surveillance Protocol Coverage Explains Unreached Cases:** Under the evaluated second-day protocol, 63.0% of incident cases develop onset after the monitoring window terminates, preventing multi-hour warning.

4.1 Distinguishing Care-Process Forecasting from Sepsis Anticipation

Forecasting antibiotic initiation and anticipating sepsis represent distinct prediction tasks. When benchmarks predict antibiotic initiation across all patients, they evaluate care-process forecasting (predicting clinician ordering behavior, including prophylactic prescribing in non-septic patients) rather than physiological sepsis anticipation. For the sepsis-anticipation estimand considered here, evaluations retain the physiological sepsis outcome while evaluating eligibility relative to clinical-process anchors; replacing the disease outcome with treatment receipt evaluates an entirely different estimand.

Clinician-vignette studies address when clinicians judge treatment appropriate [12]; our analysis addresses predictive performance relative to operational EHR proxies. These are complementary questions, and proxy precedence does not establish clinical appropriateness. Furthermore, while changes under exploratory outcome-stratified hybrid restriction were small in this cohort, differences in outcome construction, population selection, timestamp precision, and clinical-action definitions preclude direct replication of Dutta et al.'s order-based sensitivity analysis [18]. Recently in this journal, van der Vegt et al. [28] demonstrated that conventional discrimination metrics fail to account for alert timing relative to clinical events and proposed balancing predictive accuracy, alert burden, and clinical utility across decision thresholds. Clinical utility ultimately depends on alert acceptance, review feasibility, and actionable treatments [13, 15]; our simulated evaluation burden

(NNE) does not measure clinician adoption or therapeutic efficacy directly [16].

4.2 Limitations

The analysis estimates operating characteristics in a selected ICU cohort rather than clinical effectiveness. Conditioning on LOS $\geq$ 30h at surveillance entry (t = 19h) retrospectively conditions on 11 hours of future ICU survival and stay; furthermore, excluding patients developing sepsis prior to surveillance entry ($0 < T_{S3} \leq 19$h, $N = 1,478$) restricts findings to mid-stay surveillance rather than early triage. Additionally, differential surveillance lengths between right-censored covered sepsis cases (mean 7.87 hourly bins, vs 10.47 across all sepsis stays) and controls (mean 11.91 bins across all controls; 12 bins for uncensored controls) can introduce multiple-testing duration bias under peak-score aggregation ($\max_t s_i(t)$), as characterized by Do et al. [19]. Burden-matched cutoffs were derived retrospectively from evaluated score trajectories, and identifier-based selection was required for tied boundary scores; the frontier does not represent a prospectively locked policy or guarantee a specified alert rate in online streaming deployment.

Temporal precision also constrains pre-treatment opportunity assessment: while MetaVision records minute-resolved intravenous infusions in `inputevents_mv`, legacy sepsis benchmarks standardly extract `prescriptions.startdate` to capture ward and oral orders. Among the 151 surveillance-covered sepsis stays, this prescription proxy yields 0 definitely eligible, 47 indeterminate, and 104 definitely ineligible stays ($[0.0\%, 31.1\%]$ within-covered opportunity bounds; $[0.0\%, 11.5\%]$ cohort-wide). Because we assigned date-coarsened values to midnight for the primary point analysis (a_i = 00:00:00), pre-treatment opportunity cannot be definitively confirmed without minute-level electronic medication administration records (eMAR). The study did not measure clinician response, review duration, treatment changes, alarm fatigue, or patient outcomes; NNE therefore represents a simulated evaluation count rather than measured clinical workload. Alarm-reduction interventions may suppress both nonactionable and clinically important signals; our evaluation interval cannot establish the safety or optimal duration of refractory suppression during sustained surveillance [16]. Furthermore, our benchmark evaluated continuous multivariate ML models and omitted discrete integer scores (qSOFA, NEWS2), whose tie-dominated step functions resist continuous budget tuning, reflecting scope boundaries rather than ML superiority. Finally, our findings possess two distinct epistemic dimensions: the floor-discretization proof ($G_i = 24\text{h} + h \cdot \text{frac}(x_i) < 25\text{h}$) and timestamp-coarsening collapse ($N_+^{\text{Rx}} = 0$) are exact, assumption-explicit algebraic identities that hold identically in any healthcare system applying discrete lookback flooring and midnight date truncation; conversely, the empirical detection yields (1.72%) and alarm burdens reflect this specific cohort, EHR era transition (CareVue to MetaVision), and protocol schedule, where multi-center studies remain necessary to establish their operational prevalence.

5 Conclusion

In this restricted ICU cohort, observed detection yield depended on surveillance coverage, baseline discrimination, and emitted-alert policies. Evaluation feasibility was highly anchor-dependent: horizon-positive observations were retained before the selected suspicion anchor, but none remained before the earliest prescription-start proxy under midnight assignment, precluding discrimination estimation for that task. Early-warning evaluations must transparently identify outcome-dependent evaluation rules, report anchor-dependent evaluability, and benchmark under explicit workload constraints.

Ethics Approval and Consent to Participate: This study was conducted using de-identified data from the MIMIC-III (v1.4) database, collected at Beth Israel Deaconess Medical Center (BIDMC). Data collection was approved by the Institutional Review Boards of BIDMC and MIT. The data were de-identified in accordance with HIPAA Safe Harbor standards; informed consent was waived by reviewing IRBs. Access was granted under PhysioNet Credentialed DUA. **Competing Interests:** The authors declare no competing interests. **Funding:** None. **Authors' Contributions:** AS conceived the study, developed the framework, implemented pipelines, performed experiments, and drafted the manuscript. SA provided academic supervision, contributed to methodological design, critically revised the manuscript, and approved the final draft.

AI Assistance Disclosure: Large language model and AI interfaces (Perplexity AI and Anthropic/Google AI) were used for peer-review pre-submission auditing, consistency cross-checking, text refinement, and LaTeX typesetting. The authors independently designed and executed all empirical pipelines, models, and bootstrap evaluations, taking full responsibility for integrity and conclusions.

Code and Data Availability: Model architectures and frozen weight tensors are archived under analytical commit `b0932ca`; evaluation pipelines under commit `1292103`; denominator recomputation under commit `311cea3`; prior audit packaging under commit `8f10435`; and the synchronized submission manuscript release under tag `v1.0.1-freeze`. Preprocessing and evaluation pipelines are available under MIT License at `https://github.com/alireza/Sepsis-Onset-Timing-Paradox`. De-identified data are available from PhysioNet under MIMIC-III DUA.

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